

Akshath Reddy Yennam

github.com/Akshath47 | linkedin.com/in/akshathyennam | akshathreddy47@gmail.com | akshathyennam.com

EDUCATION

2024 - 2027 BEng Computing at **Imperial College London** (First Class in Year 1)
2022 - 2024 A Levels at Sancta Maria International School (Valedictorian, 4 A*, 1 A)

WORK EXPERIENCE

ValueLabs, *Software Engineering Intern* Jul 2025 – Sep 2025

- Rebuilt core components of a JD-to-resume semantic search platform using Qdrant and Node.js backend services, improving system throughput and reducing latency by $15\times$ ($90s \rightarrow 6s$)
- Developed a real time interaction service, for an AI voice interviewer, with clean modular interfaces, deterministic state transitions, and robust state management,
- Engineered high performance streaming pipelines, integrating ASR with efficient buffering, concurrency control, and predictable execution under load.
- Developed a full-stack RAG meeting assistant tool using a custom retrieval backend, improving query handling, scoring logic, and in-session state tracking.

Pulpoo, *Backend Engineer (Part-time)* Jan 2026 – Present

- Built core backend systems for a real-time AI customer support platform used by Samsung Mexico, including a scalable omnichannel memory layer for persistent user context
- Reduced latency in live interactions by restructuring backend pipelines and decoupling time-critical operations
- Developed evaluation and observability tooling to track agent performance, latency, and cost, improving system reliability

ValueLabs, *Software Engineering Intern* Jul 2023 – Aug 2023

- Built end-to-end data processing and analytics components, implementing regression, clustering, and classification modules within production-oriented Python services.
- Developed an image classification service using CNNs, packaging preprocessing, inference, and evaluation into reusable components.

PROJECTS

PintOS - Operating System

- Implemented core OS functionality in PintOS to support user processes and virtual memory, working with kernel threads, page tables, and low-level C data structures.
- Added user-level virtual memory support through demand paging and page replacement, handling page faults and maintaining supplemental page metadata to manage memory under constrained resources.

Market Time-Series Forecaster

- Designed an end-to-end financial time-series forecasting pipeline using RNN/LSTM models, with explicit handling of non-stationarity and strict separation between training and evaluation.
- Implemented rolling-window validation and regime-based analysis to prevent data leakage and evaluate model behaviour across changing market conditions.
- Analysed failure cases where performance degraded under regime changes and high noise, showing why point forecasts alone are insufficient for robust decision-making in financial time series.

Deep Research Agent

- Designed and built a distributed multi-agent research workflow that synthesizes reports, with orchestrated task handlers, concurrency control, and modular execution stages. specialized sub-agents.
- Implemented parallelized pipelines and a shared virtual filesystem, improving research throughput by $5\times$ and enabling structured fact-checked synthesis.

SKILLS & CERTIFICATIONS

Programming Languages	Python, JavaScript, Kotlin, Java, C, HTML/CSS, Haskell
Technical/Frameworks	Node.js, PostgreSQL, Docker, Git, REST APIs, Linux, Concurrency, Tensorflow, Langgraph, RAG
Certificates	Intro to Langgraph, MITx Introduction to CS, Harvard CS50